

THOMAS COATES

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Technical Skills

Languages: Python, Java, TypeScript, JavaScript, C, C++, SQL

Frameworks & Libraries: React, Angular, RxJS, Node.js

Platforms & Tools: PostgreSQL, Docker, Git, GitHub Actions, REST APIs, Supabase, MapLibre

Experience

Founder & Software Engineer

Jun 2026 – Present

TCoates LLC

Marathon, FL

- Founded a software company building custom SaaS and automation solutions for commercial clients.
- Build and operate cloud-hosted automation and data-pipeline systems in production, owning architecture, deployment, and ongoing support.
- Secured and delivered the company's first paying software contract, negotiating scope, IP/licensing terms, and pricing as a solo operator.

Software Engineer Intern

Jun 2024 – Aug 2024

ICR

Louisville, CO

- Built three reusable Angular/TypeScript components, including a configurable table system, applying SOLID principles to cut duplicate UI code and enable reuse across multiple application views.
- Implemented reactive form handling and real-time data flows using RxJS, enabling dynamic UI updates and efficient state synchronization.
- Integrated a MapLibre component into an Angular/RxJS app for real-time geospatial visualization and layer control.

MV-22 Avionics Quality Assurance Representative (CDQAR)

Nov 2013 – May 2019

United States Marine Corps

MCAS Miramar, CA

- Held Collateral Duty QA Representative authority, certifying MV-22 avionics and electrical work as flight-ready; led and mentored a team of 30+ Marines.
- Installed, inspected, and repaired MV-22 communication, navigation, and electrical systems, sustaining 80%+ readiness across a 13-aircraft fleet through shore and shipboard deployment operations.
- Served as Embarkation Manager, coordinating equipment and personnel logistics for deployments aboard USS New Orleans and USS Essex.
- Held a U.S. Government Secret security clearance (now inactive).

Projects

HailHawk | *Python, Supabase (PostgreSQL), GitHub Actions, NOAA SPC API, US Census API, Resend*

Jun 2026

- Built an automated hail-damage detection system for a roofing contractor that cross-references NOAA hail data against their customer database to surface storm-damaged properties.
- Automated data ingestion and email alerting with GitHub Actions and Resend, running unattended every 3 hours with no manual operation.
- Geocoded 2,000+ customer addresses via the US Census API, caching results to avoid redundant lookups and match storm reports to properties.
- Licensed and deployed the system to the contractor as its first paying customer, operating in production through peak hail season.
- Extending into a multi-tenant SaaS platform (Supabase, PostgreSQL, auth) to onboard additional contractors, in active development.

Cyber-Physical Control System | *Raspberry Pi, Node-RED, OpenPLC, Modbus TCP*

Apr 2025

- Built a cyber-physical control system on a Raspberry Pi 3B+ with GPIO sensors and Node-RED control logic, driving a motion-triggered, relay-controlled fan with real-time monitoring.
- Integrated OpenPLC and Modbus TCP to model SCADA-style industrial control communication.
- Analyzed Modbus TCP for security weaknesses and documented potential ICS/SCADA exploit paths.

Procedural Dungeon Generator | *Godot, GDScript, Blender*

Mar 2025

- Built a procedural cave-exploration game in Godot using modular Blender assets on a voxel grid.
- Implemented A* pathfinding with baked navigation meshes for enemy AI that pursues the player after item pickup.
- Owned asset creation, physics interactions, and system-driven level design to improve game feel.

Education

Colorado School of Mines

Aug 2025

Master of Science in Computer Science

Golden, CO

Colorado School of Mines

Dec 2024

Bachelor of Science in Computer Science

Golden, CO